

Shape Memory Alloy Actuators and Origami Construction in Soft Robotics

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Abstract—This paper discusses the applications of Shape Memory Alloys as well as other common actuation methods used within soft robotics such as pneumatics. Also discussed are the applications of origami within this field, used for encoding specific behaviours for both soft and hard-bodied robots.

I. BACKGROUND

A. Shape Memory Alloy Overview

Shape Memory Alloys (SMA) are a commonly used material throughout the field of soft robotics due to their unique behaviour among materials. They are able to be deformed and with the application of heat, will return to a predetermined shape. This can make them very useful for a wide variety of applications, for example as actuators for soft grippers.

There are many variations of alloys which exhibit the Shape Memory Effect (SME). Metallic SMAs can be broadly split into 6 groups, each one based on a main element: Nickel Titanium (Nitinol), Copper, Silver, Iron, Cobalt, Gold. These all display unique SME properties which can be altered by modifying the chemical composition, however only two display particularly useful properties, namely Nitinol and Copper based SMAs [1]. Of these two Nitinol has become the industry standard SMA for a variety of applications ranging from biomedical to aerospace. It has a higher strain recovery (8% when compared to copper) and is highly biocompatible [2], making it ideal for many medical applications such as implanting stents.

B. Phase Transition

The shape memory effect is caused by a phase transition of the material, in this case meaning a change in the structure of the atoms within the shape memory alloy. This phase transition is dependent on the combination of the two factors: heat and stress. As previously mentioned, when cold, the SMAs can be deformed, but when heat is applied it will return to its original, un-deformed, shape. This occurs due to a transition between the phases known as martensite, which is stable at cold temperatures and austenite, which is stable at high temperatures.

Figure 1 shows a strain-stress-temperature graph which details the SMA phase transition leading to the shape memory effect. Starting under complete rest and at low temperature, the material of the SMA is referred to as being in the twinned martensite phase (2). Then when stress is applied, still at low temperature the martensite becomes de-twinned (3), remaining in this phase even after stress is no longer applied (4). Once

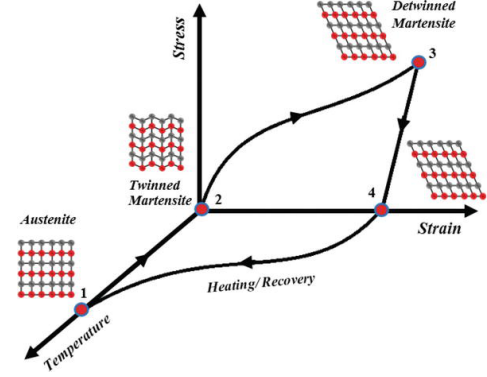


Fig. 1: SMA Phase Diagram [3]

the stress-free de-twinned martensite is heated, it becomes austenite (1) which, once cooled, transitions once again to twinned martensite (2), recovering the original shape [4].

C. Pseudoelasticity

Another useful property of shape memory alloys is so called Superelasticity or Pseudoelasticity. This occurs during the austenite phase (above the so-called Austenite finish temperature, denoted A_f . Table II) and allows the material to deform even further, incurring little to no plastic deformation. While above A_f and under no load, the SMA is in a austenite cubic crystal structure and while loaded the structure changes microscopically from cubic to de-twinned martensite [1]. Then once unloaded the crystal structure changes back to austenite.

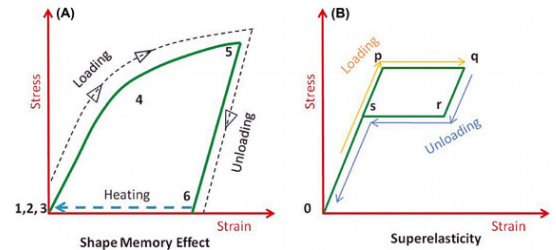


Fig. 2: Pseudoelasticity Phase Diagram [5]

Figure 2 shows a comparison of the shape memory effect and pseudoelasticity on stress-strain graphs. Here the first graph shows the cycle explained in Section I-B. The second graph occurs only at point (1) in Figure 1 past the A_f temperature threshold.

D. Nitinol Properties

One of the most useful properties of SMAs from an actuation perspective is that they have a very high power to weight ratio. Table I shows the performance data of Nitinol as an actuator in comparison with some other actuators commonly used for soft robotics.

TABLE I: Performance Comparison of Actuator Types [6]

Parameter	Ni-Ti SMA	Hydraulic	Pneumatic
Stress (MPa)	200	20	0.7
Strain (%)	10	50	50
Efficiency (%)	3	80	90
Bandwidth (Hz)	3	4	20
Work per Volume (J/cm^3)	10	5	0.175
Power per Volume (W/cm^3)	30	20	3.5

It shows that the power to weight ratio (in this case measured as MPa) is significantly higher than that of hydraulic and pneumatic actuators. It does however only deform by approximately 10% in comparison to 50% for the other actuation types and is much less energy efficient, primarily due to heat loss. It does have a higher work and energy per volume metric due to a large density of stored energy. In contrast hydraulics require large fluid volumes in order to achieve high forces and pneumatics uses air, which has a low energy density [6]. Despite this, most other methods of actuation are able to produce much higher raw force than Nitinol and other SMAs, making them more suited for smaller scale applications.

TABLE II: Material properties of Nitinol [7]

Martensite Young's modulus (E_m)	23.2 GPa
Austenite Young's modulus (E_a)	49.5 GPa
Martensite start temperature (M_s)	77 °C
Martensite finish temperature (M_f)	42 °C
Austenite start temperature (A_s)	81 °C
Austenite finish temperature (A_f)	106 °C

Table II shows some of the material properties of common Nitinol, although variations can be made to work within different temperature ranges.

II. APPLICATIONS OF SMAS IN SOFT ROBOTICS

A. SMAs as Actuators

One of the most common applications of SMAs, such as Nitinol is for use as an actuator. This typically involves Nitinol wires, referred to as tendons, embedded within a polymer matrix to provide structure and faster stroke recovery. The tendons constrict when heat is applied by passing a current through the wire. This alleviates one of the issues presented by alternative actuation methods, which is that they often require large or cumbersome external elements such as hydraulics or motors.

One of the main drawbacks to the efficacy of SMA tendons is that the force they are able to apply is heavily dependent on

the length of the wire, with longer tendons able to apply more force. One paper proposes a solution to this, by encapsulating the Nitinol tendons in silicon rubber tubing, allowing them to freely slide. This decouples the length of the tendon and the design of the polymeric matrix allowing for a much higher actuation force than the matrix would otherwise allow [8].

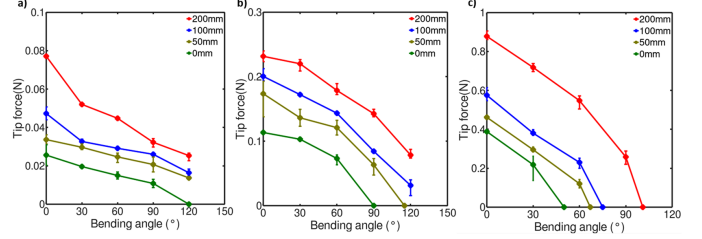


Fig. 3: Tip force for different tendon lengths of actuators with a polymeric matrix thickness of (a) 3 mm, (b) 5 mm, and (c) 8 mm. The different colours represent the length of wire outside of the matrix (halved as the wire doubles back on itself) [8].

Figure 3 shows that there was a noticeable increase in the tip force of the soft gripper, the longer the total length of the tendon. This does however incur other problems, as the external length of the tendon needs to be stored somewhere. This will likely involve routing the wire around various joints, which may lead to reductions in efficiency and compactness.

Another approach for increasing the force of SMAs is a combined approach with another actuator. One paper proposes a hybrid DC motor - SMA humanoid hand actuator, where the motor drives one or multiple tendons which incorporate SMA wires. Principle Component Analysis (PCA) is then used to decouple the various actuators dependant on the desired pose of the hand. It was found that this analysis could increase the synergic strength of the actuator significantly [9]. The SMA wires were attached directly to the pulleys of the motors and pass through a series of electrical contacts at exponentially increasing lengths along the wire. This format of segmentation along the SMA is known as Segmented Binary Control [10]. This method allows for several moving components per axis. The combination of SMAs and DC motors with PCA, allows for the entire hand to be driven using only two motors [9].

Pneumatics are common place throughout soft robotics due to high force output. One such actuator is the Pneumatic Actuated Muscle (PAM) which consists of a hollow flexible tube, surrounded by a braided mesh. When pressurised the tube expands in width, but contracts in length due to the restriction of the mesh [11]. This principle can be expanded upon by utilising SMAs in order to strategically restrict the tube's expansion along specific axes by selectively activating the SMA tendons. One paper experimented with this principle and the results showed that this method produced precise control of the PAM in multiple axes through the precise actuation of different SMA coils [12].

In the context of pneumatics, SMAs can also be used as an actuator in order to compress a volume of air. A study was conducted, using SMA coils to compress air within a chamber, by using a heating element to activate the SMAs. This has the benefit of increasing the air pressure due to thermal expansion, in addition to the compression provided by the SMA coils [13]. While this effect was minimal, the study demonstrates the potential use of SMAs in order to compress air, which removes the need for a large air compressor. This concept would however, only be applicable for small scale

B. SMAs for Locomotion

Another use for the properties of shape memory alloys is for locomotion. The following paper produced a small, soft robot with movement produced by mimicking the locomotion patterns of inchworms [14]. Figure 4 shows the locomotion principle of the robot

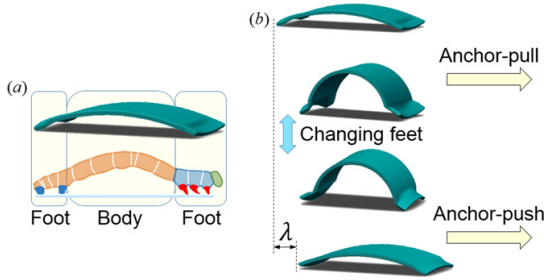


Fig. 4: Locomotion principle of the inchworm robot [14]

This behaviour of the inchworm is mimicked by the robot by the cyclic activation of the SMA tendons. Longitudinal muscles embedded in the body are used to provide the deformation of the body, pulling or pushing the legs. Each side can be activated asymmetrically in order to produce a turning motion. The activation and deactivation of the legs is produced by the front and back tendons which run perpendicular to the length of the body. When activated, the body deforms placing the low-friction feet in contact with the floor and removing the high-friction PDMS. This allows the legs to be slid, while the other end of the robot remains in place. When all SMAs are activated in the correct cycle (allowing 15 seconds between activations for the SMAs to cool), locomotion is produced [14].

III. ORIGAMI IN SOFT ROBOTICS

A. Origami for Encoding behaviour

Another widely researched topic, with a large number of potential applications throughout many different industries is origami. Different designs can be used in order to encode a wide variety of behaviours. This behaviour can be quite complex an example of which can be found in the paper 'Origami mechanologic'. The study utilises a material known as PEDOT:PSS, which responds to relative humidity by the absorption and desorption of water, which leads to a swelling and shrinking of the material. This material was used in an origami 'waterbomb' type fold in order to encode logic

gates entirely mechanically [15]. The topic of origami logic gates has also been studied further in the paper 'Origami Logic Gates for Printable Robots', which developed bistable switches made from flexible polyester film. These were able to switch between two different circuits to power low current LEDs. These circuits do have the drawback that due to high resistances only low currents can be effectively utilised, posing challenges for direct use with shape memory alloy actuators [16].

B. Origami for Enabling Movement

Another paper uses concepts from origami to create a fully untethered soft robot, capable of exhibiting many different behaviours for a variety of applications. The origami body (shown in Figure 5) has a rectangular cross-section with a fully folded length of about 40mm and a width of approximately 30mm. It uses a polyethylene terephthalate (PET) film in order to give it large compressive and bending deformations, while the origami structure provides high torsional resistance, making it ideal for this application. Shape memory alloy coils are used as actuators for this robot [17].

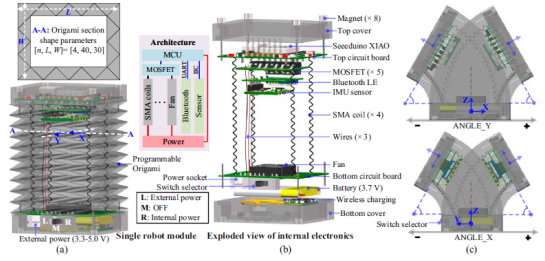


Fig. 5: Origami untethered soft robot design [17]

Particularly impressive is the fact that all the components were able to fit inside the body, including a small fan for temperature regulation of the SMAs and an Internal Measurement Unit (IMU) for more accurate control. The effectiveness of the IMU was measured using LEDs on the top of the robot and two external cameras for x and y angles. The results show a clear and consistent measurement of the robot state.

Different actuation methods have also been applied to moving origami robots. One such method is through magnetic actuation. The following paper was able to control the movement of an origami robot by manipulating the magnitude and direction of an external magnetic field. The robot itself consisted of four Kresling origami patterns, arranged in two-level symmetry shown in Figure 6. These Kresling folds can be compressed by their rotation which is translated into lateral motion through the use of high and low friction points of contact with the floor. While this is similar to the inchworm robot discussed previously, the points of high and low friction are encoded through the shape of the feet, rather than their material [18].

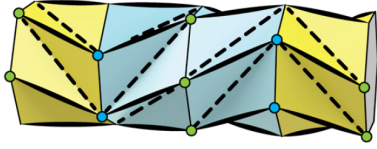


Fig. 6: Origami crawler design. Each of the four segments is a Kresling origami fold arranged in a two-level symmetric pattern [18]

The Kresling folds can be made in multiple different ways, varying parameters such as the number of folds (in this case a hexagonal cross section, or 6 folds) which can lead to differing characteristics. A study found six fold symmetry was able to produce the highest recoil percentage, when testing Kresling structures with 4, 5, 6, 7 and 8 folds [19].

Origami robots have also been actuated using pneumatics. A paper entitled: 'Soft pneumatic actuator-driven origami-inspired modular robotic *pneumagami*' explored this method of actuation using Soft Pneumatic Pouch Actuators [20]. This method of actuation is itself a wide field with countless different designs such as McKibben actuators or PAMs mentioned in Section II-A. They have a wide variety of applications such as wearables with pneumatic actuation for elderly care [21]. The Pneumagami example uses these actuators to control configurable modules with 3 Degrees of Freedom (DoF) which can be stacked on top of each other to modify their functionality. It uses the 'waterbomb' origami pattern a spherical joint with 2D rigid plate structures connected by hinges. In this case the origami structure provides the stiffness to constrain the torsional motion which is undesirable.

Inspiration from origami structures can also be applied to the internal structure of robots as well. The following paper developed an endoskeleton design based on the Kresling fold. This allowed it to utilise the Kresling's ability to fold down lengthways into a much smaller structure. This exoskeleton was then covered in a soft skin allowing it to be actuated by pneumatics [22]. When compared to other soft pneumatic actuators such as McKibben actuators, this allows for a much more compact design when negative air pressure is applied.

IV. RESEARCH GAPS

Many of the technologies mentioned previously such as SMAs, pneumatics or origami are well studied for applications within many different fields. However, there are significantly fewer studies documenting the results of integrating such technologies together. Soft robotics as a new field presents a unique opportunity for such studies to be performed and applied to a wide variety of fields ranging from medical to industrial. This paper has already highlighted some of the studies on these topics, but despite the large amounts of research on different types of actuator methods or robot designs there remains huge potential for new and innovative

TABLE III: Prototype Maximum Retraction Length (mm)

Stacked Segments	1	2	3	4
Full Extension (Resting State)	36	67	100	129
Size after Full Retraction	23	43	65	87
Percent of Full Size	64	64	65	67

research.

With regards to utilising origami for further research there is an almost endless possibility for new shapes and designs as well as different materials to choose from. More than just providing the structure of a robot, origami can be used in combination with different material properties in order to provide complex behaviour patterns depending on its actuation as well as environmental stimulus [15].

There are also further opportunities for origami structures to be used in conjunction with different methods of actuation. While there have been some studies utilising origami with magnetic [18], pneumatic [22] or SMA [17] actuators, there are still many possibilities for further research using hybrid actuation methods or different origami folds. Furthermore, while complex behaviour encoding structures, such as logic gates [16] [15], have been studied in isolation, there are few papers exploring these concepts directly for soft robotics applications.

While such applications may seem niche they can provide significant contribution to many different fields. There are already studies that propose using origami micro-robots with magnetic actuation, as seen with the Kresling-fold crawler [18], in order to enter specific areas of the body in order to deliver liquid medicine [23]. This demonstrates the potential applications of such technologies, and why research into these fields is so important.

V. RESULTS

A. Vertical Compression

The following section details the results and properties of the constructed prototype. The design consists of modular Kresling sections made from 0.5mm sheets of laser cut polypropylene which can be stacked on top of each other. Tendons running through the actuator are spooled up and down by 5v 28BYJ-48 stepper motors, which control the motion of the device. In practice however, these motors proved to be underpowered and while control was possible it was limited by the use of these motors. The NEMA17 stepper motor might have been a more appropriate choice in this instance due to a significantly higher torque.

Table III shows the effective range of vertical compression of the stacked Kresling units. They are able to be compressed by approximately one third of their total size which remains roughly consistent across the number of segments. Strong and consistent values relating to vertical compression along its z axis demonstrates the success of the prototype in its key selling point as an actuator capable of compressing its own volume. Alongside minimising its total size, a further benefit is that it

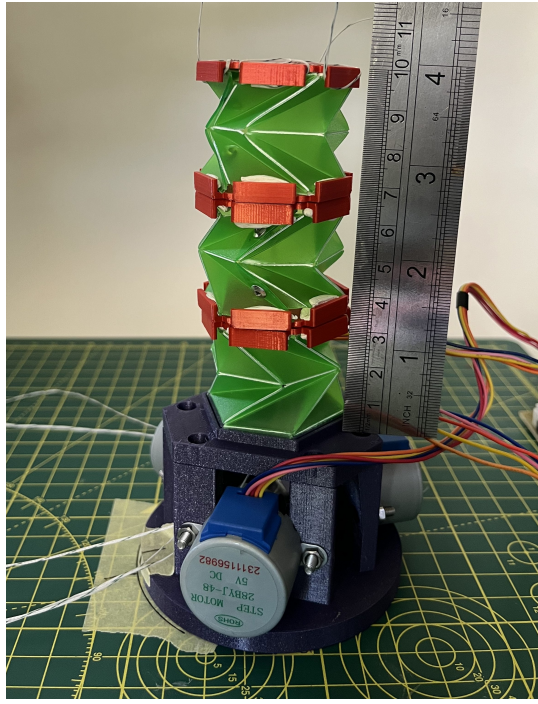


Fig. 7: Experiment Setup with 3 Kresling Sections

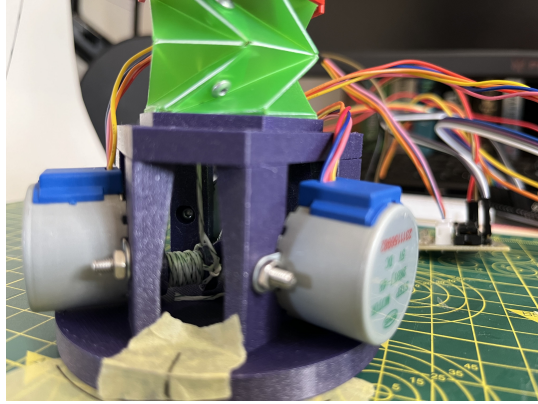


Fig. 8: Tendon Spool

is able to reach points more easily than more standard rigid body actuators by compressing its own volume, where a rigid actuator would rely on extending its joints sideways to lower the vertical position of the end-effector.

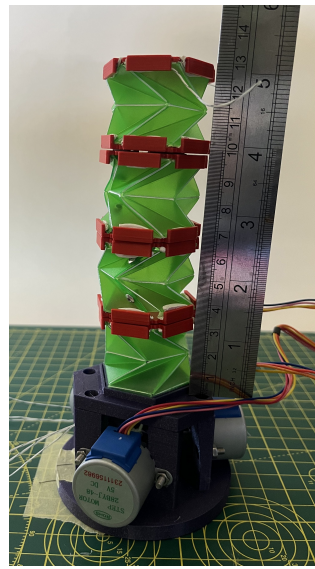
B. Horizontal Control

C. Motor Control

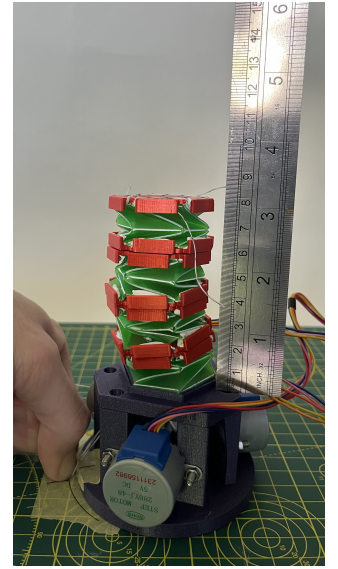
As previously mentioned, the chosen motors were overall underpowered and so were not able to provide the full range

TABLE IV: Horizontal bend length (mm) per segment

Segments	1	2	3	4
Horizontal Bend	3	11	28	36

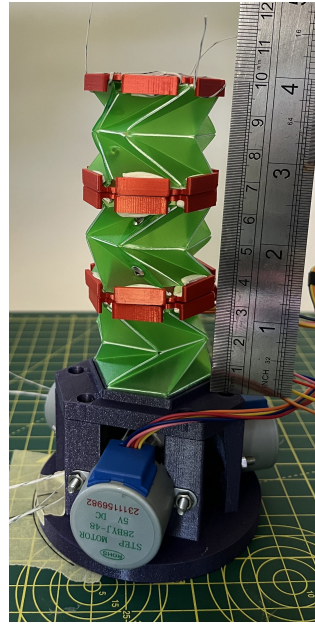


(a) At rest/ full extension

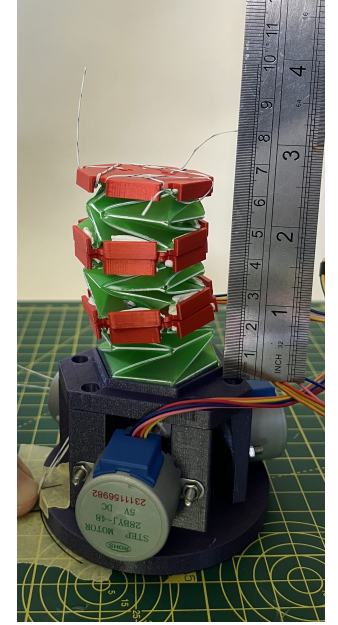


(b) At full vertical compression

Fig. 9: Experiment showing full vertical compression using 4 Kresling segments



(a) At rest/ full extension



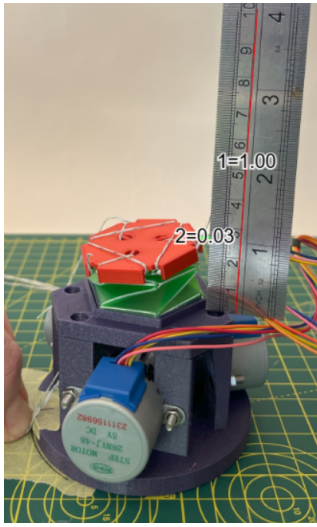
(b) At full vertical compression

Fig. 10: Experiment showing full vertical compression using 3 Kresling segments

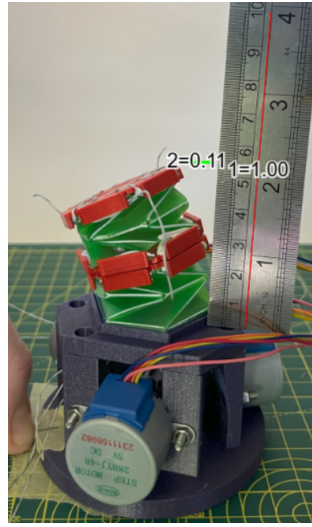
of motion seen in the results. Despite this control was still possible and effective utilising the motors.

D. Conclusion

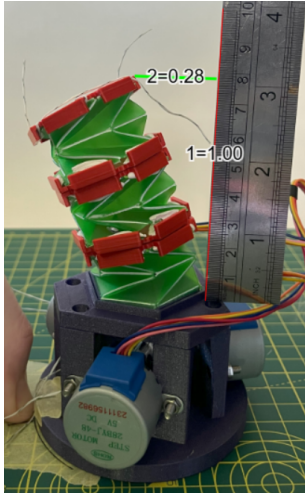
This section has detailed the successful implementation of a Kresling-fold origami actuator, able to perform vertical compression without any horizontal expansion. Despite some downsides such as weak motors, the design performed as



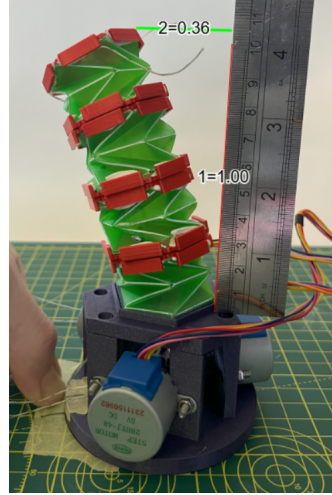
(a) 1 Kresling Core



(b) 2 Kresling Cores



(c) 3 Kresling Cores



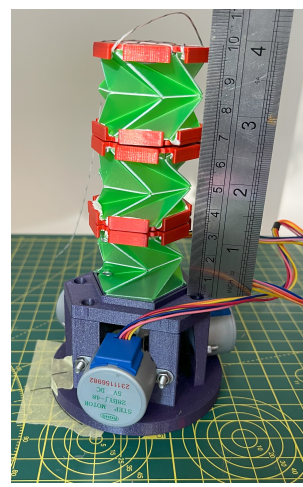
(d) 4 Kresling Cores

Fig. 11: Comparison of the horizontal bending motion of the actuator

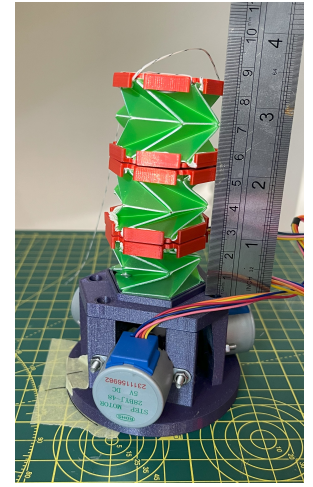
expected and was able to demonstrate an effective range of motion. It also showed how the design could be easily modified by adding or removing Kresling sections. Currently the design uses the natural stiffness of the polypropylene to provide the expansion once the motors unwind the tendons. There is potential to expand the design further by testing other materials which could provide greater flexibility, resulting in an improved range of motion.

VI. CONCLUSION

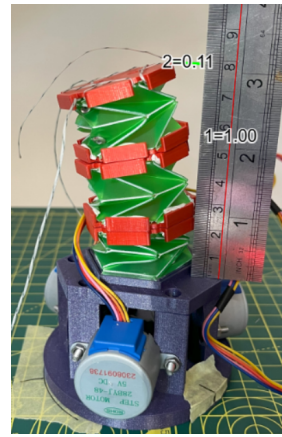
The research presented in this literature review has demonstrated the applications and effectiveness of a variety of topics being actively researched in the soft robotics community. While there is much current active research in the fields of shape memory alloys and origami applications for engineering, many opportunities for further research in these fields still



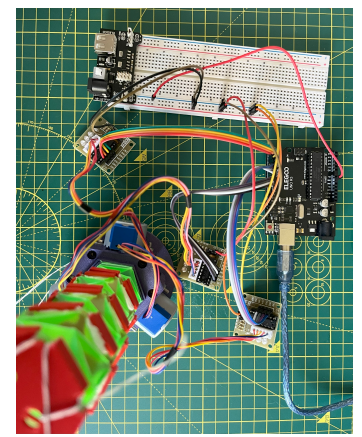
(a) Full Extension



(b) Vertical Compression



(c) Horizontal Motion



(d) Circuit

Fig. 12: Circuit and resultant actuator motion

remain. There is therefore, an argument for future research on these topics which may lead to further synergic discoveries regarding these technologies and new, unseen applications may reveal themselves. The design proposed in the latter section of the report will provide a basis for the study of using origami structures to encode different behaviours for pneumatic-SMA driven actuators.

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